AUDITMOS

Secure your space

STARGATE.FINANCE

audit / code review report

July 30, 2025

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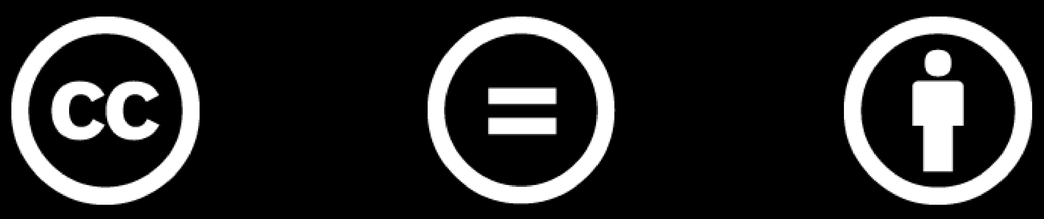
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APPROACH AND METHODOLOGY

PURPOSE

1. Determine the correct operation of the protocol, according to the design specification.
2. Identify possible vulnerabilities that could be exploited by an attacker.
3. Detect errors in the smart contract that could lead to unexpected behavior.
4. Analyze whether best practices were followed during development.
5. Make recommendations to improve security and code readability.

CODEBASE

GitHub	https://github.com/piotr-layerzero/locker/tree/main
Branch	main
Contracts	*

METHODOLOGY

1. Reading the available documentation and understanding the code.
2. Doing automated code analysis and reviewing dependencies.
3. Checking manually source code line by line for security vulnerabilities.
4. Following guildlines and recommendations.
5. Preparing this report.

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DESCRIPTION

Issues Categories:

Severity	Description
CRITICAL	vulnerability that can lead to loss of funds, failure to recover blocked funds, or catastrophic denial of service.
HIGH	vulnerability that can lead to incorrect contract state or unpredictable operation of the contract.
MEDIUM	failure to adhere to best practices, incorrect usage of primitives, without major impact on security.
LOW	recommendations or potential optimizations which can lead to better user experience or readability.

Each issue can be in the following state:

State	Description
PENDING	still waiting for resolving
ACKNOWLEDGED	know but not planned to resolve for some reasons
RESOLVED	fixed and deployed

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AUDIT SCOPE

1. getting to know the project	✓
2. research into architecture	✓
3. manual code read	✓
4. permissions of state changing functions	✓
5. identify common Solidity vulnerabilities	✓
6. test coverage	✓
7. static analysis	✓
8. storage key overlaps	✓
9. DOS possibilities by malicious attacker	✓
10. steal funds possibilities	✓

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FINDINGS

Finding	Severity	Status
#1 - Misleading comment	LOW	ACKNOWLEDGED
#2 - Emit an event in crucial places	LOW	ACKNOWLEDGED
#3 - Redundant error	LOW	ACKNOWLEDGED

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#1 - MISLEADING COMMENT

Severity	Status
LOW	ACKNOWLEDGED

The comment above the beneficiary variable incorrectly states that only this address can release funds:

```
// only this address can release funds
address private immutable beneficiary;
```

However, no access control is implemented in the contract, any address can call withdraw() or release(). This makes the comment misleading and potentially confusing to future readers or auditors.

RECOMMENDATION

Update the comment to accurately reflect that any address can trigger fund release, but the funds are always sent to the beneficiary.

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#2 - EMIT AN EVENT IN CRUCIAL PLACES

Severity	Status
LOW	ACKNOWLEDGED

The TimeLocker contract does not emit events in key functions such as withdraw() and release(). Emitting events in these functions is important for off-chain tracking, transparency and enabling dApps to monitor token movements.

RECOMMENDATION

Emit events whenever funds are transferred from the contract.

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#3 - REDUNDANT ERROR

Severity	Status
LOW	ACKNOWLEDGED

The custom error TimeLocker__NotBeneficiary is declared in the TimeLocker contract but is never used anywhere in the code.

```
error TimeLocker__NotBeneficiary(); // Declared but unused
```

RECOMMENDATION

Remove the redundant error.

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